

Matthieu Vinet

Robotics Engineering Student — Polytech Sorbonne



Born 10 Jan 2004

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Seeking a 6-month final-year internship · Robotics · Paris · 2027

Final-year robotics student at Polytech Sorbonne, back from an R&D internship in Berlin where I built the vision-based precision docking of an autonomous mobile robot. Comfortable across the whole stack — ROS2, vision, electronics, mechanics — and most interested in what it takes to make a robot work reliably outside the lab, not just in a demo.

EDUCATION

Engineering Degree (MEng), Robotics · Polytech Sorbonne 2024 – 2027
Paris, France

Engineering Preparatory Cycle (PeiP) · Polytech Tours 2022 – 2024
Tours, France

Scientific Baccalaureate (high-school diploma) · Lycée Isaac de l'Étoile 2019 – 2022
Poitiers, France — majors in Mathematics & Engineering Sciences, advanced Maths option

EXPERIENCE

Robotics Engineering Intern · Botshare.ai Apr. 2026 – Jul. 2026
Berlin, Germany — on-site (Mechatronics Lab, BHT University)

- Recommissioned the robot, brought up autonomous navigation, then designed its precision docking system for wireless charging — camera + a bundle of 3 AprilTags, visual servoing, 6-phase docking sequence (open-source OpenAMRobot project).
- Worked the failure modes rather than the happy path: reliable docking under changing lighting and with people moving through the scene.
- Developed and documented the `openamrobot_docking` ROS2 package (Gazebo simulation, Nav2 integration) ; merged PRs on the `openamr-platform-sw` and `-fw` repositories.
- **Published a technical article** on the Botshare.ai blog explaining the engineering approach and why robustness beats raw precision — botshare.blog

Member then Treasurer · Robotech Association (robotics — Polytech Sorbonne) Since Sep. 2024

- Member since 2024 and treasurer for one year (April 2025 – April 2026, handover completed); active in the club's revival and in several team projects (mobile robot, arcade cabinet, RC plane, DIY motor).

Other · Freelance: design and deploy showcase websites for craftspeople (HTML/CSS/JS, Cloudflare Pages) ; wwoofing on a Norwegian farm (summer 2025, daily English) ; crew member at McDonald's (2022) ; volunteer at the SPA animal shelter (2023).

PROJECTS

SO-101 (open-source arm, 5 DoF + gripper, The Robot Studio): 3D-printed, assembled and calibrated the arm (6 Feetech servos) ; set up reliable leader/follower teleoperation with LeRobot, as a platform for imitation-learning experiments (dataset collection and policy evaluation in progress).

Self-balancing inverted pendulum (personal project, ongoing): two-wheeled robot stabilised by a cascaded controller at 200 Hz on ESP32 ; steady-state angle error cut from 0.293° to 0.064° — by proving the first architecture could not work, then tracing the remaining error to a bug in the manufacturer's I2C driver. C++, MPU-6050, NEMA 17 / A4988.

AGRI-BOT (industrial project, ROB4): autonomous mobile robot for remote supervision of Polytech Sorbonne's connected greenhouse — RTK GPS navigation, LiDAR, teleoperation and real-time video streaming, web control HMI. Team project, defended April 2026.

Electronics & fabrication: DIY brushless motor, arcade cabinet, RC plane, ESP-NOW remote-controlled robot — prototyping at the FabLab (3D printing, laser cutting).

Software: pan-tilt camera with gesture recognition (OpenCV) ; car counting by image processing (C++/OpenCV) ; mini shell in C (fork, pipes, redirections) ; chess AI in C ; poker AI (Monte-Carlo, Python).

→ All my projects in detail at matthieu-vinet.fr

TECHNICAL SKILLS

Robotics: ROS2, Nav2, Gazebo, micro-ROS, OpenCV, visual servoing, AprilTag, SLAM, EKF ; sim-to-real debugging on physical robots

AI & machine learning: imitation learning with LeRobot (SO-101 teleoperation pipeline) ; reinforcement learning — Sarsa(λ) with linear approximation, Q-learning (Easy21)

Programming: Python, C, C++, JavaScript, HTML / CSS ; Git, Docker, Linux ; daily use of AI coding assistants

Electronics & embedded: breadboard prototyping, soldering, component selection, power electronics (DC-DC), sensors (LiDAR, IMU), DC & brushless motors (BLDC) ; Arduino, ESP32, Raspberry Pi boards

CAD & prototyping: SolidWorks, 3D printing, laser cutting (FabLab), Matlab

LANGUAGES

French — native · English — professional

INTERESTS

Video games · Programming · Prototyping & robotics

SOFT SKILLS

Curious · Rigorous · Self-driven · Fast learner